

UMP Tests for one-sided composite vs. one-sided composite:

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Para Inf.

Let $X \sim f(x|\theta) : \theta \in \Theta \subset \mathbb{R}$ Test $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$

Def: A family $\{f(x|\theta) : \theta \in \Theta \subset \mathbb{R}\}$ is said to have a Monotone Likelihood Ratio (MLR) property in some real-valued statistic $T(x)$ if for any $\theta_1 < \theta_2$ ($\theta_1, \theta_2 \in \Theta$)

$\frac{f(x|\theta_2)}{f(x|\theta_1)}$ is non-decreasing in $T(x)$

for all x for which either $f(x|\theta_2) > 0$ or $f(x|\theta_1) > 0$

i.e. $\forall x \in \{x: f(x|\theta_1) > 0\} \cup \{x: f(x|\theta_2) > 0\}$

(Recall: $\frac{c}{0} = \infty$)

Note: (1) In def. assumed for $\theta_1 \neq \theta_2$, $f(x|\theta_1) \neq f(x|\theta_2)$

(2) Typically $T(x)$ is considered to be a sufficient stat for θ

(3) What if $\frac{f(x|\theta_2)}{f(x|\theta_1)} \downarrow T(x)$?

Then $\frac{f(x|\theta_2)}{f(x|\theta_1)}$ is $\uparrow$ in $(-T(x))$ or $\frac{1}{T(x)}$. Hence, no need to define MLR property using non-increasingness

- Ex: (1) $N(\mu, \sigma_0^2)$ (2) $N(\mu_0, \sigma^2)$ (3) $\text{Bin}(n, p)$ (4) $\text{Pois}(\lambda)$
 $\nearrow$ known $\nearrow$ known
(5) $U(0, \theta)$ (6) $U(\theta, \theta+1)$ (7) $\text{Laplace}(\theta, \sigma)$ with either θ or σ known
(8) $\text{Logistic}(\theta, 1)$

General one-parameter exponential family:

Suppose $f(x|\theta) = h(x) e^{\eta(\theta)T(x) - B(\theta)}$, $\theta \in \mathbb{R}$ and $\eta(\theta)$ is strictly increasing.

For $\theta_1 < \theta_2$

$$\frac{f(x|\theta_2)}{f(x|\theta_1)} = e^{(\eta(\theta_2) - \eta(\theta_1))T(x) - (B(\theta_2) - B(\theta_1))}$$

Let's consider (1) $X \sim U(0, \theta)$ Let $\theta_1 < \theta_2$

$$\frac{f(x|\theta_2)}{f(x|\theta_1)} = \begin{cases} \frac{\theta_1}{\theta_2} & \text{if } 0 \leq x < \theta_1 \\ \infty & \text{if } \theta_1 \leq x < \theta_2 \end{cases}$$

(2) $X \sim U(\theta, \theta+1)$ Let $\theta_1 < \theta_2$

$$\frac{f(x|\theta_2)}{f(x|\theta_1)} = \begin{cases} 0 & \text{if } \theta_1 < x < \theta_2 \\ 1 & \text{if } \theta_2 < x < \theta_1+1 \\ \infty & \text{if } \theta_1+1 < x < \theta_2+1 \end{cases}$$

(Case 1)

$$\frac{f(x|\theta_2)}{f(x|\theta_1)} = \begin{cases} 0 & \text{if } \theta_1 < x < \theta_1+1 \\ \infty & \text{if } \theta_2 < x < \theta_2+1 \end{cases}$$

(Case 2)

Non-example: ① Cauchy($\theta, 1$)

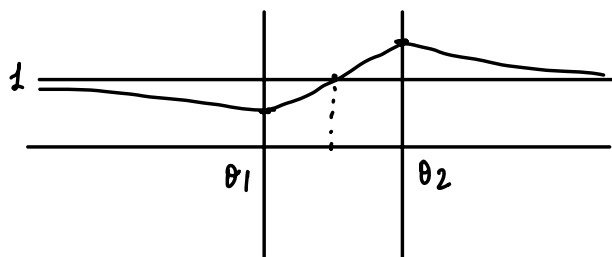
$$f(x|\theta) = \frac{1}{\pi(1+(x-\theta)^2)} \quad \text{Let } \theta_1 < \theta_2$$

$$h(x) = \frac{f(x|\theta_2)}{f(x|\theta_1)} = \frac{1 + (x - \theta_2)^2}{1 + (x - \theta_1)^2}$$

$$x = \theta_1 \quad h(x) < 1$$

$$x = \theta_2 \quad h(x) > 1$$

$$x \rightarrow \pm\infty \quad h(x) \rightarrow 1$$



$\Rightarrow \{f(x|\theta): \theta \in \mathbb{R}\}$ is not MLR in x

② $N(\theta, \theta^2)$, $\theta \in \mathbb{R}$ not MLR in x (H.W.)

• Benefit of MLR property:

Lemma: If $X \sim f_\theta(x)$ and $\{f_\theta(x): \theta \in \Theta\} \subseteq \mathbb{R}$ has MLR in $T(x)$

and if ψ is a non-decreasing fn. of $T(x)$, then $g(\theta) = E_\theta[\psi(T(X))]$ is non-decreasing in θ

Pf: Take $\theta_1 < \theta_2$ Define $A = \{x: \frac{f_{\theta_2}(x)}{f_{\theta_1}(x)} < 1\}$

$$B = \{x: \frac{f_{\theta_2}(x)}{f_{\theta_1}(x)} > 1\}$$

Define

$$a := \sup_{x \in A} \psi(T(x))$$

$$b := \inf_{x \in B} \psi(T(x))$$

Claim: $b \geq a$

Pf: $h(T(x)) > 1$

$$\Rightarrow T(x) \geq h^{-1}(1) = c_1$$

$$:= \inf\{y: h(y) \geq 1\}$$

$$B \subseteq \{x: T(x) \geq c_1\} \subseteq \{x: \psi(T(x)) \geq \psi(c_1)\}$$

$$b = \inf_{x \in B} \psi(T(x)) \geq \psi(c_1)$$

$$\frac{f_{\theta_2}(x)}{f_{\theta_1}(x)} = h(T(x))$$

$$\begin{aligned}
A &= \{x: h(T(x)) < 1\} && \text{if } h(y) < 1 \\
&\subseteq \{x: T(x) \leq c_1\} && \Rightarrow y \leq h^{-1}(1) = c_1 \\
&\subseteq \{x: \psi(T(x)) \leq \psi(c_1)\} && \text{True b/c } y > c_1 \Rightarrow h(y) \geq 1
\end{aligned}$$

$$a = \sup_{x \in A} \psi(T(x)) \leq \psi(c_1) = b$$

$$\begin{aligned}
g(\theta_2) - g(\theta_1) &:= \int \psi(T(x)) f_{\theta_2}(x) dx - \int \psi(T(x)) f_{\theta_1}(x) dx \\
&= \int \psi(T(x)) (f_{\theta_2}(x) - f_{\theta_1}(x)) dx
\end{aligned}$$

$$\int \underbrace{\psi(T(x))}_{\leq a} \underbrace{(f_{\theta_2}(x) - f_{\theta_1}(x))}_{\leq 0} dx + \int \underbrace{\psi(T(x))}_{\geq b} \underbrace{(f_{\theta_2}(x) - f_{\theta_1}(x))}_{> 0} dx + 0$$

$$\geq a \int_A (f_{\theta_2}(x) - f_{\theta_1}(x)) dx + b \int_B (f_{\theta_2}(x) - f_{\theta_1}(x)) dx$$

$$= a \underbrace{\left(\int_{\mathbb{R}^n} (f_{\theta_2}(x) - f_{\theta_1}(x)) dx - \int_{A^c=B} (f_{\theta_2}(x) - f_{\theta_1}(x)) dx \right)}_0 + b \int_B (f_{\theta_2}(x) - f_{\theta_1}(x)) dx$$

$$= (b-a) \int_B (f_{\theta_2}(x) - f_{\theta_1}(x)) dx > 0$$

Karlin-Rubin Theorem

Suppose $X \sim f_{\theta}(x)$ and $\{f_{\theta}(x): \theta \in \Theta \subseteq \mathbb{R}\}$ has MLR in $T(x)$

Consider Testing $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$ (θ_0 known)

(i) For prefixed $0 < \alpha < 1$, there exists a UMP size- α test given by

$$\phi^*(x) = \begin{cases} 1 & \text{if } T(x) > c \\ \gamma & \text{if } T(x) = c \\ 0 & \text{if } T(x) < c \end{cases} \quad \text{where } c \geq 0 \text{ and } \gamma \in [0, 1] \text{ are determined from}$$

$$\beta_{\phi^*}(\theta_0) = E_{\theta_0} \phi^* = \alpha$$

(ii) The power $\beta_{\phi^*}(\theta)$ strictly increasing for all θ for which $0 < \beta_{\phi^*}(\theta) < 1$

(iii) For $\theta < \theta_0$, ϕ^* minimizes $\beta_{\phi}(\theta)$ for any ϕ satisfying $\beta_{\phi}(\theta_0) = \alpha$

i.e. $\beta_{\phi^*}(\theta) \leq \beta_{\phi}(\theta) \quad \forall \quad \beta_{\phi}(\theta_0) = \alpha$

(iv) Assume $P_{\theta}(k_{\theta}(x) = K k_{\theta_0}(x)) = 0 \quad \forall \theta > \theta_0$ and for all $K \geq 0$

If a test ϕ satisfies $\beta_{\phi}(\theta_0) = \beta_{\phi^*}(\theta_0) (= \alpha)$

For $\theta > \theta_0$, then either $\beta_{\phi^*}(\theta) > \beta_{\phi}(\theta)$ or $\phi^*(x) = \phi(x) \quad P_{\theta} \text{ a.e.}$

$$(P(\phi^*(X) = \phi(X)) = 1 \quad \forall \theta > \theta_0)$$

(v) Fix $\theta_1 \in \Theta_1$. Consider testing $H'_0: \theta \leq \theta_1$ vs. $H'_1: \theta > \theta_1$
Then ϕ^* is UMP size- α_1 test where $\alpha_1 := \beta_{\phi^*}(\theta_1)$

Implication:

Ex 1. $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$

$H_0: \mu \leq 0$ vs. $H_1: \mu > 0$

$$\phi^*(x) = \mathbb{1}(\sqrt{n}\bar{x} > z_\alpha)$$

$H'_0: \mu \leq 1$ vs. $H_1: \mu > 1$ — ②

ϕ^* is UMP test for ② with size $= \beta_{\phi^*}(1)$

Ex 2. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{exp}(\lambda)$ $f(x|\lambda) = \lambda e^{-\lambda x} \mathbb{1}(x > 0)$

$$f(x|\lambda) = \lambda^n e^{-\lambda \sum_1^n x_i} \quad \text{Fix } \lambda_2 > \lambda_1$$

$$\frac{f(x|\lambda_2)}{f(x|\lambda_1)} = \left(\frac{\lambda_2}{\lambda_1}\right)^n e^{-\sum x_i (\lambda_2 - \lambda_1)} \downarrow T(x) = \sum_1^n x_i$$

UMP size- α Test: $H_0: \lambda \leq \lambda_0$ vs. $H_1: \lambda > \lambda_0$

By Karlin-Rubin: UMP test given by $\phi(x) = \begin{cases} 1 & \text{if } \sum x_i < c \\ 0 & \text{if } \sum x_i > c \end{cases}$

$$\text{Find } c: \mathbb{P}_{\lambda_0}(\sum_1^n X_i < c) = \alpha$$

$$\Rightarrow \mathbb{P}_{\lambda_0}(\underbrace{2\lambda_0 \sum_1^n X_i}_{\chi^2_{2n}} < \underbrace{2\lambda_0 c}_{\chi^2_{2n, 1-\alpha}}) = \alpha$$

$$X \sim \text{exp}(\lambda) \\ 2\lambda X \sim \chi^2_2$$

$$\Rightarrow \boxed{2\lambda_0 c = \chi^2_{2n, 1-\alpha}}$$

$\therefore$ UMP size- α test $\phi(x) = \mathbb{1}(2\lambda_0 \sum_1^n x_i < \chi^2_{2n, 1-\alpha})$

$$\begin{aligned} \beta_{\phi}(\lambda) &= \mathbb{P}_{\lambda}(2\lambda_0 \sum X_i < \chi^2_{2n, 1-\alpha}) \\ &= \mathbb{P}_{\lambda}(2\lambda \sum X_i < \frac{\lambda}{\lambda_0} \chi^2_{2n, 1-\alpha}) \\ &= F_{\chi^2_{2n}}\left(\frac{\lambda}{\lambda_0} \chi^2_{2n, 1-\alpha}\right) \end{aligned}$$